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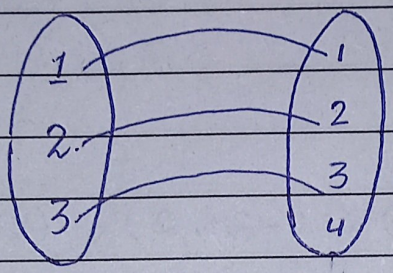
Topic # 01

"Functions"

* **Domain**: Input is domain
* **Range**: Images (output) are range
* **Co-Domain**: - combining both -



Example:-

1. $y = x^2$ Find domain and range

Domain = Real number $\mathbb{R}$ or $(-\infty, \infty)$

Range = $[0, \infty]$

2. $h(x) = \frac{1}{x+1}$

Domain = $(-\infty, -1) \cup (-1, \infty)$

Range = $R - \{0\}$

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3. $f(x) = \sqrt{x-5}$

Domain = $(+5, \infty)$

Range = $(0, \infty)$

4. $g(x) = \sqrt{4x^2-1}$

$4x^2 - 1 \geq 0$

$4x^2 \geq 1$

$x^2 \geq \frac{1}{4}$

$x \geq \pm \frac{1}{4}$

$(-\infty, -1/2) \cup (1/2, \infty)$ → Domain!

5. $\frac{1}{x^2-4}$

Domain = $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

* INTERVALS :-

$\{x \in R / a < x < b\}$

- Open Interval

(a, b)

- Semi open Interval

$[a, b) = \{x \in R / a \leq x < b\}$

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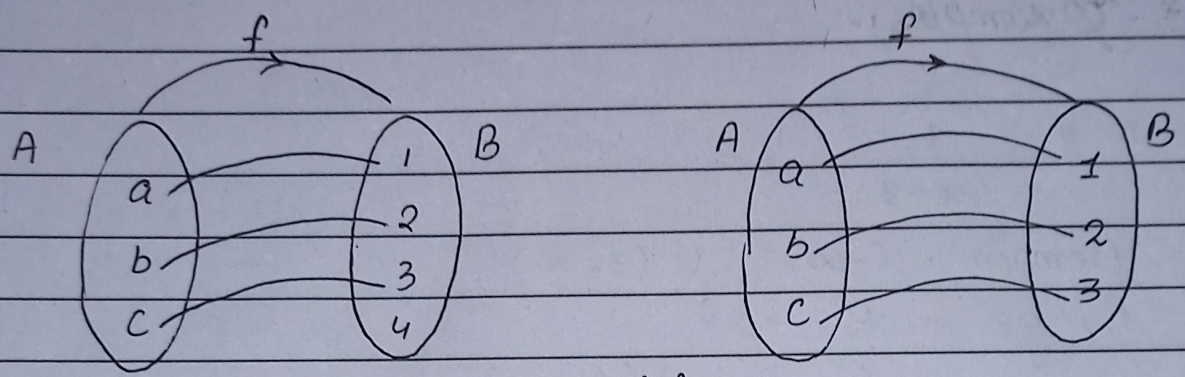
- Close Interval

$[a, b] = \{x \in R / a \leq x \leq b\}$

Types Of Functions :-

1) One-to-One Function (injective function)

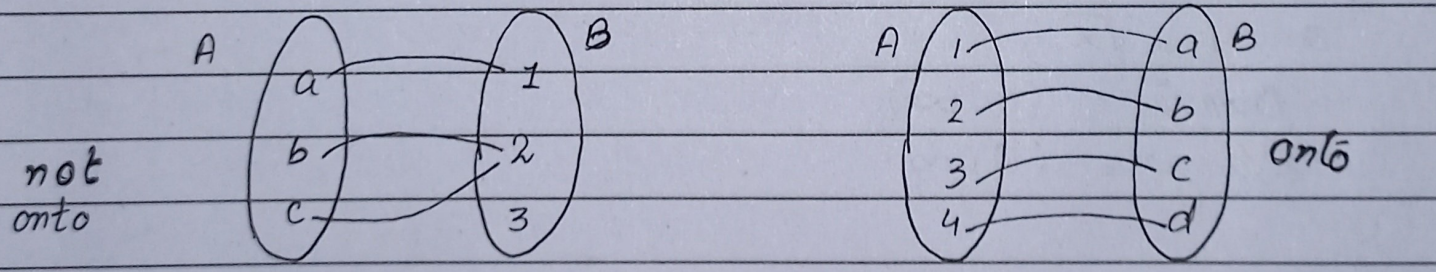
A function $f: A \to B$ is one-to-one if distinct element of set A has distinct images in set B.



no-repetition

2) Onto-function (sujective)

A function $f: A \to B$ is onto function if the range of A is B that is co-domain is equal to range.



co-domain = {a, b, c, d}

range = {a, b, c, d}

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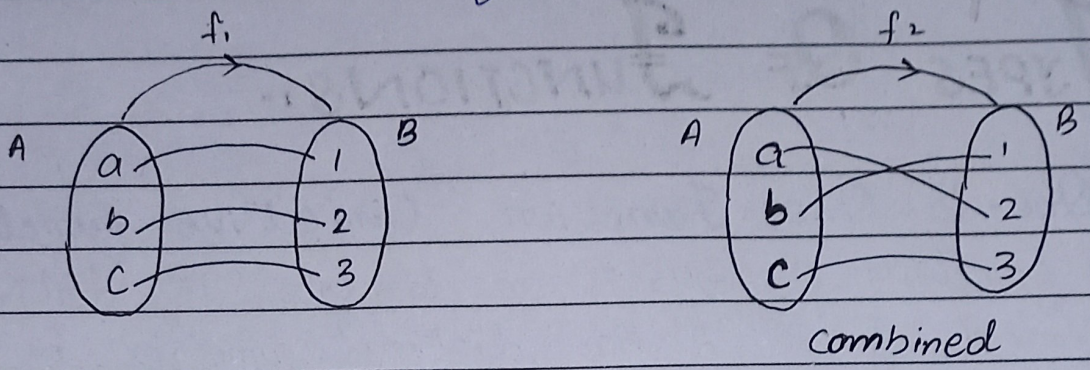
3) One-One & Onto function (Bijective)

If the function has the property of both one-one and onto function.



* Examples :-

1. $y = \frac{1}{x-3}$

Domain = $(-\infty, 3) \cup (3, \infty)$

Range = $R - \{0\}$

= $(-\infty, 0) \cup (0, \infty)$

2. $y = \frac{1}{x^2-9}$

Domain = $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

Range = $(-\infty, 0) \cup (0, \infty)$

3. $f = \sqrt{x}$

Domain = $(0, \infty)$

Range = $(0, \infty)$

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4) $f(x) = \sqrt{x+3}$

Domain = $(-3, \infty)$

Range = $(0, \infty)$

5) $f(x) = \frac{1}{\sqrt{25-x}}$

Domain = $(25, -\infty)$

Range = $(0, \infty)$

6) $f(x) = \sqrt{x^2-4}$

Domain = $(2, \infty)$

Range = $(0, \infty)$

7) $f(x) = \sqrt{2(x-1)}$

Domain = $(1/2, \infty)$

Range = $(0, \infty)$

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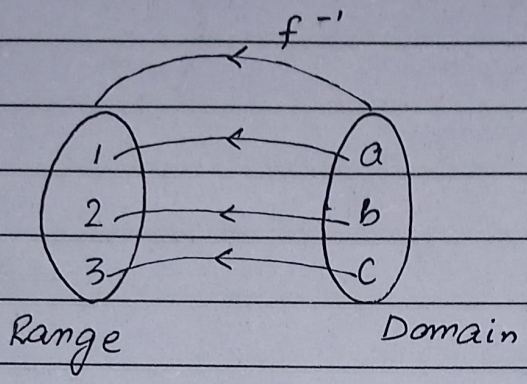
* Inverse Function:-

If $f: A \to B$ be a (one-to-one & onto) function (Bijective) function the inverse of $f \to f': B \to A$ is inverse function of f and is defined as:-

$$a = f^{-1}(b), \quad \forall\ b \in B$$



* Examples:-

1) $y = \dfrac{2x-1}{x-1}$

$$y(x-1) = 2x-1$$
$$xy - y = 2x - 1$$
$$xy - 2x = -1 + y$$
$$x(y-2) = -1 + y$$

$$x = \frac{-1+y}{y-2}$$

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2) $f(x) = \sqrt{x+9}$

$f(x) = y$

$y = \sqrt{x+9}$

$y^2 = x+9$

$y^2 - 9 = x$

$x = y^2 - 9$

* Absolute Value Function:-

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$|2| = 2$

$|-2| = -(-2) = 2$

* Composition Of Function:-

Given two functions f & g, the composile function fog (also called the composition)

- $x \rightarrow g \rightarrow g(x) \rightarrow f \rightarrow f(g(x))$
- $x \rightarrow g(x) \rightarrow \frac{1}{f(g(x))}$

} fog

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* Examples:-

if $f(x) = x^2$ & $g(x) = x - 3$
find fog and gof

* fog:

$f(x) = x^2$, $g(x) = x - 3$

$(fog)(x) = f(g(x))$

$= f(x - 3)$

$fog = (x - 3)^2$

gof:-

$gof = g(f(x))$

$= g(x^2)$

$gof = x^2 - 3$ Answer!

Que. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2 - x}$. find
(a) fog (b) gof (c) fof (d) gog

(a) fog:

$f(g(x)) = f(\sqrt{2 - x})$

$= 4\sqrt{2 - x}$

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(b) gof.

$$g(f(x)) = g(\sqrt{x})$$
$$= \sqrt{2-\sqrt{x}}$$

(c) fof.

$$fof = f(f(x))$$
$$= f(\sqrt{x})$$
$$= \sqrt{\sqrt{x}}$$

(d) gog.

$$gog = g(g(x))$$
$$= g(\sqrt{2-x})$$
$$= \sqrt{2-\sqrt{2-x}}$$

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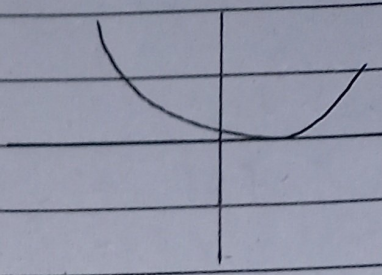
- GRAPHS -

Que. $f(x) = x^2$ (Quadratic eq.)

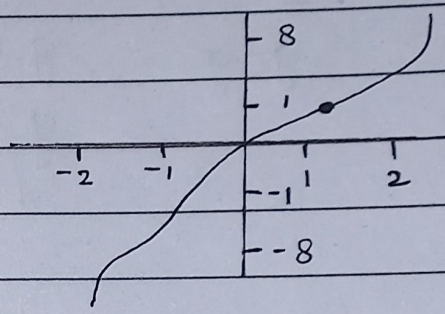
Domain = R

Range = $[0, \infty)$



Que. $f(x) = x^3$ (cubic function)

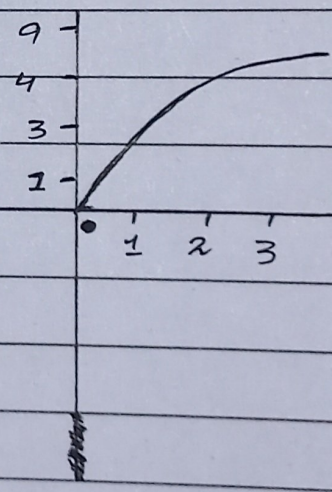
x	$f(x) = x^3$
-2	-8
-1	-1
0	0
1	1
2	8



Domain = R & Range = R

Que. $f(x) = \sqrt{x}$

x	$f(x) = \sqrt{x}$
0	0
1	1
3	1.73
4	2
9	3



Domain = $[0, \infty)$

Range = $[0, \infty)$

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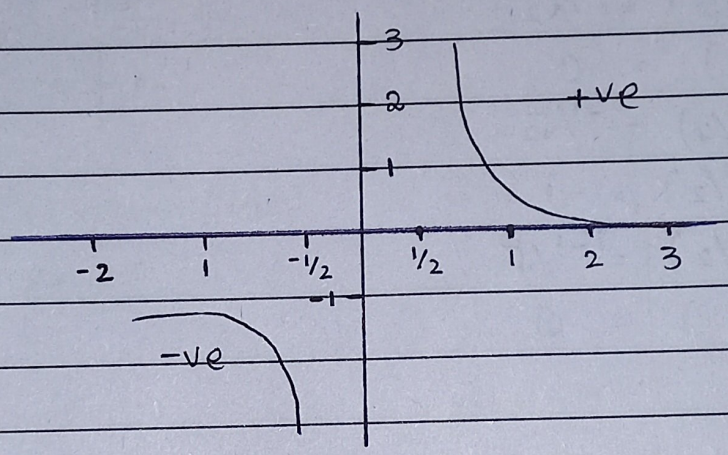
Que. $f(x) = 1/x$

x	$f(x) = 1/x$	
$1/3$	3	(+ve)
$1/2$	2	
1	1	
2	$1/2$	
3	$1/3$	

x	$f(x)$	
-3	$-1/3$	(-ve)
-2	$-1/2$	
$-1/2$	-2	
$-1/3$	-3	

Domain = $(-\infty, 0) \cup (0, \infty)$

Range = $(-\infty, 0) \cup (0, \infty)$



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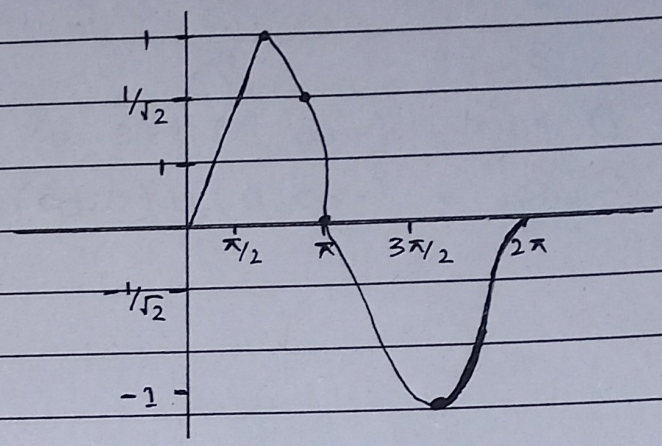
* TRIGNOMETRIC FUNCTION:-

Que $f(x) = \sin x$

Domain = R

Range = $[-1, 1]$

"x"	"$f(x) = \sin x$"
0	$\sin(0) = 0$
$\pi/4$	$\sin(\pi/4) = 1/\sqrt{2}$
$\pi/2$	$\sin(\pi/2) = 1$
$3\pi/2$	$\sin(3\pi/2) = 1/\sqrt{2}$
π	$\sin(\pi) = 0$
$5\pi/2$	$\sin(5\pi/4) = -1/\sqrt{2}$
$3\pi/2$	$\sin(3\pi/2) = -1$
$7\pi/2$	$\sin(7\pi/2) = -1/\sqrt{2}$
2π	$\sin(2\pi) = 0$

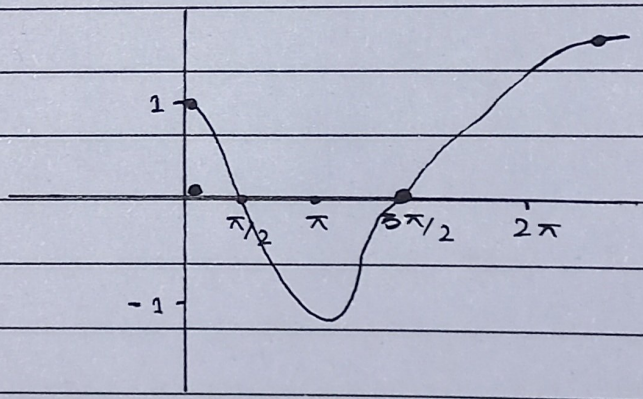


Que. $f(x) = \cos x$

Domain = R

Range = $[-1, 1]$

x	$f(x) = \cos x$
0	1
$\pi/2$	0
π	-1
$3\pi/2$	0
2π	1



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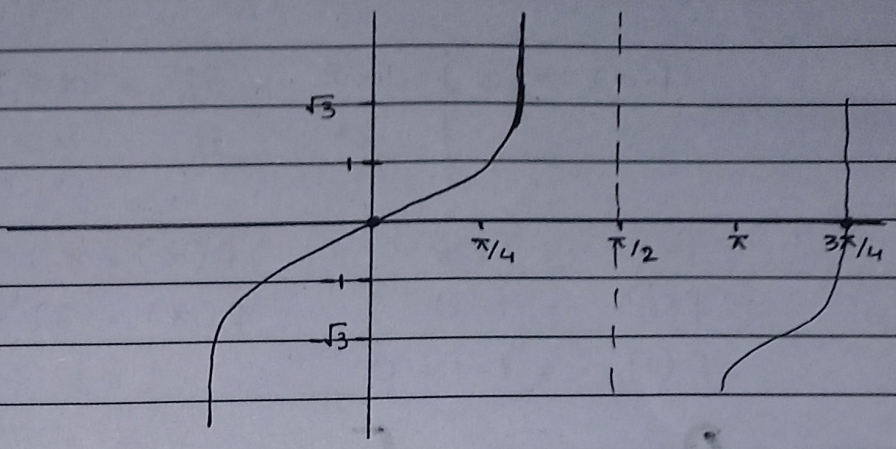
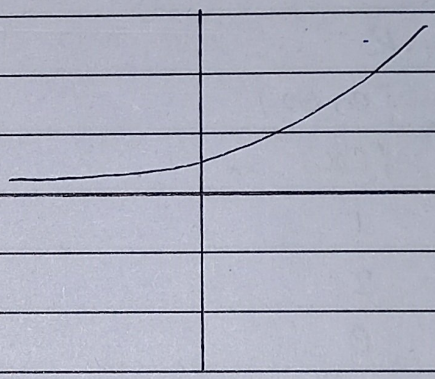
Que. $f(x) = \tan x$

x	$f(x) \tan x$
0	0
$45° = \pi/4$	1
$60° = \pi/3$	$\sqrt{3}$
$90° = \pi/2$	∞
$120° = 2\pi/3$	$-\sqrt{3}$
$135° = 3\pi/4$	-1
$180° = \pi$	0
$240° = 4\pi/3$	$\sqrt{3}$
$225° = 5\pi/4$	1
$270° = 3\pi/2$	∞

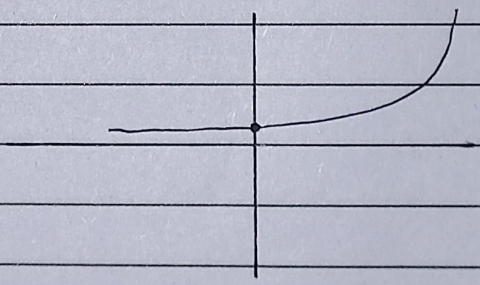


Que. $y = e^x$ (Exponential)

x	$y = e^x$
-2	
-1	$1/e$
0	$e^0 = 1$
1	e^1
2	e^2
3	e^3



Que. $f(x) = a^x$



Domain (f) = R

Range = $(0, \infty)$

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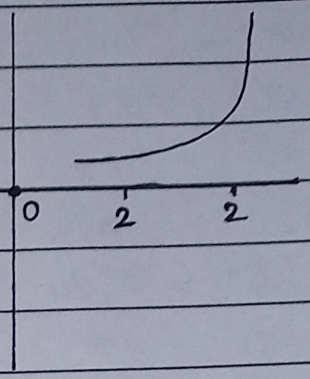
* Piece Wise Function :-

A function is defined (Peice-wise)

$$f(x) = \begin{cases} 1-x & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$



$f(x) = 1-x$ $\qquad$ $f(x) = x^2$

$f(0) = 1-0 = 1$ $\qquad$ $f(x) = (2)^2 = 4$

$f(1) = 1-1 = 0$

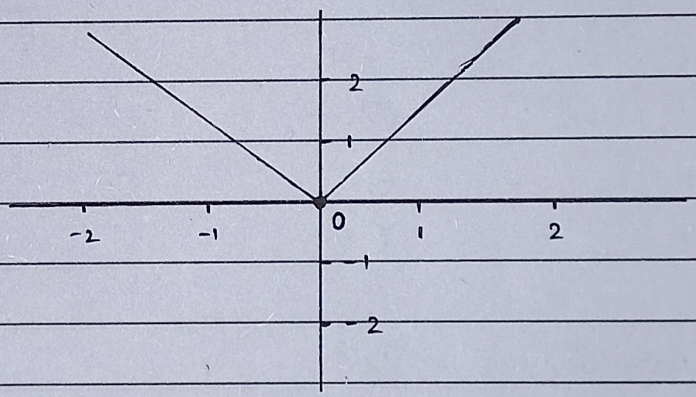
* Absolute Function :-

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Domain = R

Range = $[0, \infty)$

x	$f(x)$
-1	1
-2	2
0	0
1	1
2	2



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* Ceiling Function :- $[x]$ $\lceil x \rceil$

for $x \in R$, $[x]$ = least integer $\geq x$

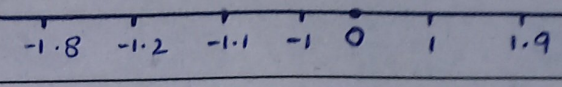
$[-1.1] = -1$

$[-1.8] = -1$

$[1.9] = 2$

$[1.1] = 2$

$z = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$



* Floor Function :- $\lfloor x \rfloor$

$[x]$ = greatest integer $\leq x$

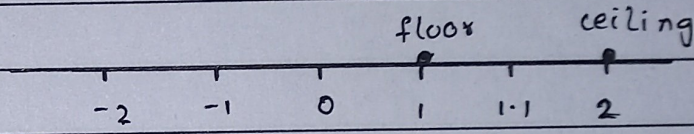
(floor of x)

Examples :-

- 1.2 $\lceil x \rceil$, $\lfloor x \rfloor$, = ?

  $\lfloor 1.2 \rfloor = 1$ (floor)

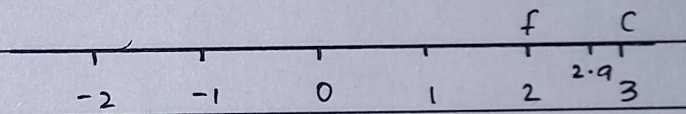
  $\lceil 1.2 \rceil = 2$ (ceiling)



- 2.9

  $\lfloor 2.9 \rfloor = 2$ (f)

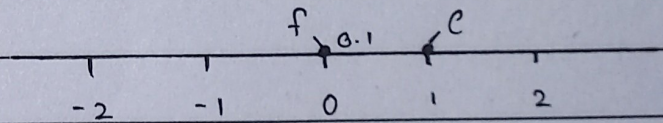
  $\lceil 2.9 \rceil = 3$ (C)



- 0.1

  $\lfloor 0.1 \rfloor = 0$ (f)

  $\lceil 0.1 \rceil = 1$ (C)



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- n is an integer

  $\lfloor n \rfloor = n \qquad \lceil n \rceil = n$

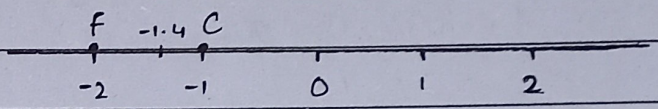
In particular we can define the floor function and ceiling function in following way.

$$x \in R, \quad n \leq x \leq n+1$$

$$\lfloor x \rfloor = n, \quad \lceil x \rceil = n+1$$

And if x is an integer, then $\lfloor x \rfloor = \lceil x \rceil = n$

- $\lfloor -1.4 \rfloor = -2 \qquad \lceil -1.4 \rceil = -1$



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Topic #02

"Limits"

* Computational Rule for Limit at a Point :-

(i) $f(x) = C$ (constant function)

$$\lim_{x \to a} f(x) = C = \lim_{x \to a^+} f(x)$$

$$\lim_{x \to a} f(x) = C$$

(ii) $f(x) = x$

$$\lim_{x \to a} x = a = \text{limit}_{x \to a^+}$$

(iii) $f(x) = x^2$

$$\text{limit}_{x \to 2a^-} x^2 = a^2 = \lim_{x \to a^+} x^2$$

$$\text{limit}_{x \to a} x^2 = a^2$$

Date: _____

* LAWS FOR LIMITS :-

Suppose c is a constant, a is a real number.

$$\lim_{x \to a} f(x) = L_1 \quad \text{(exist at a)}$$

$$\lim_{x \to a} g(x) = L_2 \quad \text{(exist at a)}$$

① $$\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} [f(x)] + \lim_{x \to a} [g(x)]$$

e.g. $f(x) = x^2$, $g(x) = x^3$

$$\lim_{x \to 2} [x^2] + \lim_{x \to 2} x^3 = 2^2 + 2^3$$

② $$\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$$

e.g. $f(x) = x^2$ $\quad g(x) = x^3$

$$\lim_{x \to 2} [x^2 - x^3] = \lim_{x \to 2} x^2 - \lim_{x \to 2} x^3$$

③ $$\lim_{x \to a} [f(x) \cdot g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$$

e.g. $f(x) = x^2$, $g(x) = x^3$

$$\lim_{x \to 2} (x^2 \cdot x^3) = \lim_{x \to 2} x^2 \cdot \lim_{x \to 2} x^3$$

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Date: _____

④ $\lim_{x \to a} [c f(x)] = c \lim_{x \to a} f(x)$

e.g. $\lim_{x \to 2} [7x^2] = 7 \lim_{x \to 2} x^2$

$= 7(2)^2$

$= 28$

⑤ $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$ $\quad \lim_{x \to a} g(x) \neq 0$

e.g. $f(x) = x^2 + 4$, $g(x) = x^3$

$\lim_{x \to 2} \frac{f(x)}{g(x)} = \frac{x^2+4}{x^3} = \frac{2^2+4}{2^3} = \frac{8}{8} = 1$

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Date:

* LIMITS OF RATIONAL NUMBERS:-

Case : 1

$\frac{P(x)}{q(x)}$, $q(x) \neq 0$ [Here $P(x)$ & $q(x)$ are Polynomials]

e.g. $f(x) = \frac{x^2-1}{x-1}$

$\lim_{x \to 0} f(x) = \frac{0^2-1}{0-1} = 1$

Case : 2

e.g. ①

$\lim_{x \to 1} \frac{x^2-1}{x-1} \longrightarrow \frac{P(x)}{q(x)}$

$\lim_{x \to 1} \frac{(x-1)(x+1)}{(x-1)}$

$(x+1) = 1+1$
$= 2$

e.g. ②

$\lim_{x \to -2} \frac{x^2+x-2}{x^2+3x+2}$

$= \frac{(x+2)(x-1)}{(x+1)(x+2)}$

Date: _____

$$\lim_{x \to -2} \frac{(x-1)}{(x+1)}$$

$$= \frac{-2-1}{-2+1}$$

$$= \frac{-3}{-1} = 3 \quad \text{Answer!}$$

Case : 3

$$\lim_{x \to 1} = \frac{x^2 + x - 2}{x^2 - 2x + 1}$$

$$= \frac{(x+2)(x-1)}{(x-1)^2}$$

$$= \frac{x+2}{x-1}$$

$$= \frac{1+2}{1-1} = \frac{3}{0} \quad (\text{infinite})$$

Homework : $\lim_{x \to 0} \frac{(x+3)^2 - 9}{x}$

$$= \frac{x^2 + 6x}{x}$$

$$= x + 6 = 0 + 6$$

$$= 6$$

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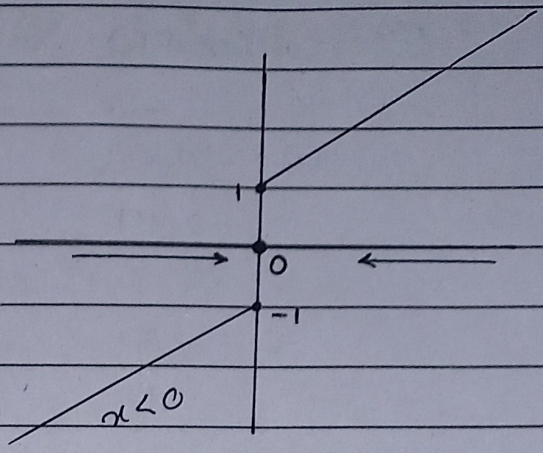
Date: _____

- limit Basic Que.

$$f(x) \begin{cases} x+1 & x \geq 0 \\ x-1 & x < 0 \end{cases}$$



$$\lim_{x \to 0^-} f(x) = -1$$

$$\lim_{x \to 0^+} f(x) = 1$$

* LEFT HAND LIMIT :-

By approaching '0' from the left, if $f(x)$ approaches to L_1 then the left hand limit of $f(x)$ at a is L_1

$$\lim_{x \to 0} f(x) = L_1$$

* RIGHT HAND LIMIT :-

By approaching 'a' from the right, if $f(x)$ approaches to L, then R.H.S limit of $f(x)$ at a is L_2

$$\lim_{x \to a^+} f(x) = L_2$$

- left hand limit is not always equal to right hand limit.

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* Existence of two sided limit at a point.

Let $y = f(x)$ be a function and $a \in R$

if $\lim_{x \to a^-} f(x) = L_1$

if $\lim_{x \to a^+} f(x) = L_2$

Non-Existence Of a Limit:-

Case : 1

$\lim_{x \to a^-} f(x) = L_1$

$\lim_{x \to a^+} f(x) = L_2$

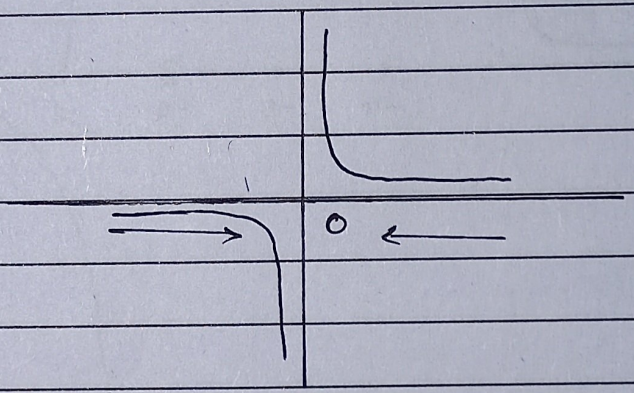
$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$

Limit does not exist!

Case : 2

$\lim_{x \to 0^+} f(x) = +\infty$

$\lim_{x \to 0^-} f(x) = -\infty$



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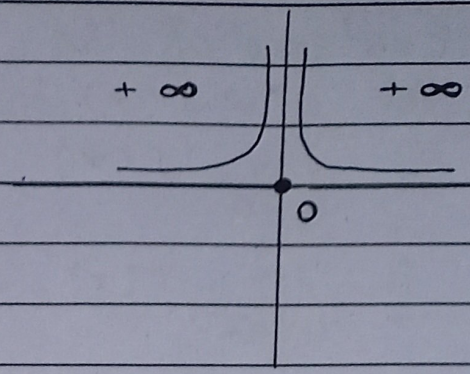
Case : 03

$$\lim_{x \to 0^+} f(x) = +\infty$$

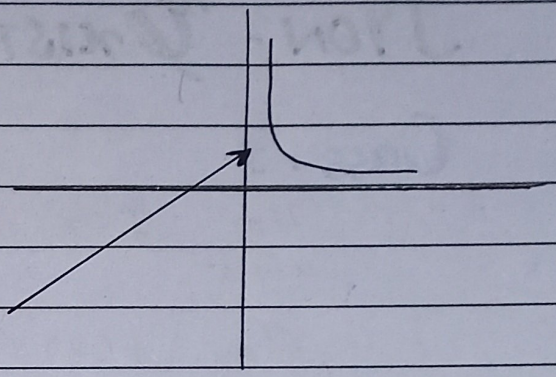
$$\lim_{x \to 0^+} f(x) = +\infty$$



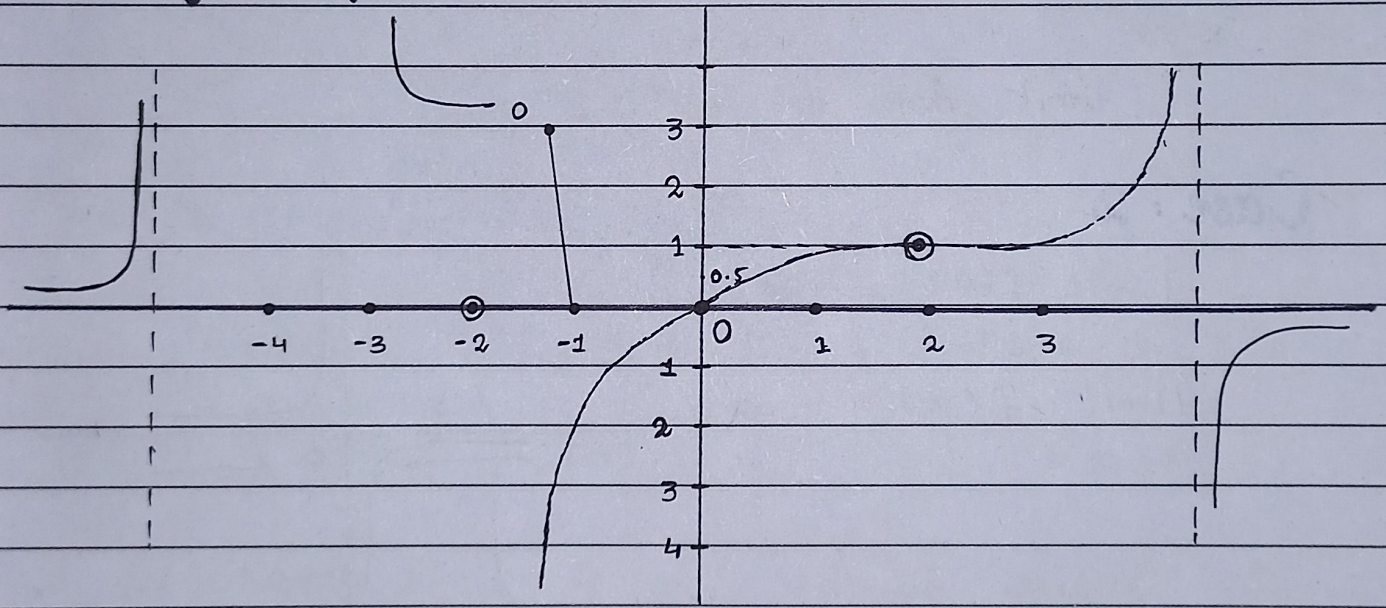
Case : 04

$$\lim_{x \to a^-} f(x) = l$$

$$\lim_{x \to a^-} f(x) = \infty$$



- By Graph -



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$\lim_{x \to 0^-} f(x) = 0$

$\lim_{x \to 0^+} f(x) = 0$

$\lim_{x \to 2^-} f(x) = 0.5$

$\lim_{x \to 2^+} f(x) = 0.5$

$\lim_{x \to 4^+} f(x) = -\infty$

$\lim_{x \to 4^-} f(x) = \infty$

$\lim_{x \to -2^-} f(x) = 3$

$\lim_{x \to -2^+} f(x) = 2$

$\lim_{x \to -4^-} f(x) = +\infty$

$\lim_{x \to -4^+} f(x) = +\infty$

MIGHTY PAPER PRODUCT

Date: _____

Que. $g(x) = \begin{cases} x+1 & x < 2 \\ 3 & x = 2 \\ 5-x & 2 < x < 5 \\ 2 & x = 5 \end{cases}$

Draw a graph for

1) $\lim_{x \to 2^-} g(x)$

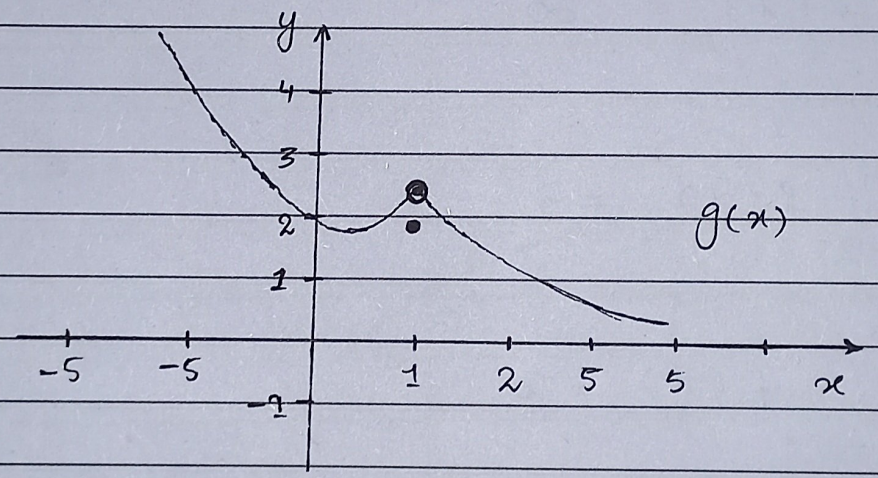
2) $\lim_{x \to 2^+} g(x)$

3) $\lim_{x \to 2} g(x)$

4) $\lim_{x \to 5^-} g(x)$

5) $\lim_{x \to 5^+} g(x)$

6) $\lim_{x \to 5} g(x)$



MIGHTY PAPER PRODUCT

Date:

1) $\lim_{x \to 2^-} g(x) = 3$

2) $\lim_{x \to 2^+} g(x) = 3$

3) $\lim_{x \to 2} g(x) = 3$

4) $\lim_{x \to 5^-} g(x) = 0$

5) $\lim_{x \to 5^+} g(x) =$ does not exist

6) $\lim_{x \to 5} g(x) =$ does not exist

MIGHTY PAPER PRODUCT

Date: _____

* Questions for Practice :-

(1) $\lim_{x \to -2} \dfrac{x^2 + x - 2}{x^2 + 3x + 2}$

$$\frac{(x-1)(x+2)}{(x+1)(x+2)}$$

$$\frac{x-1}{x+1} = \frac{-2-1}{-2+1} = 3$$

(2)

$$f(x) = \begin{cases} \dfrac{x^2-1}{x-1} & x > 1 \\ \sqrt{2x^2-1} & x \leq 1 \end{cases}$$

find $\lim_{x \to -1} f(x)$

∵ the function is piecewise, we have to find $\lim_{x \to -1} f(x)$. Since, 1st part is $(x > 1)$ we won't simplify it because -1 is not greater than 1.

So,

$$\sqrt{2x^2-1} = \sqrt{2(-1)^2-1} = \sqrt{2-1} = 1$$

MIGHTY PAPER PRODUCT

Date: _____

(3) $\lim_{x \to 0} \frac{|x|}{x}$

left hand limit $x \to 0^-$

$|x| = -x$

$\frac{-x}{x} = -1$

Right hand limit $x \to 0^+$

$|x| = x$

$\frac{x}{x} = 1$

Since, left hand limit $\neq$ right hand limit
it does not exist.

MIGHTY PAPER PRODUCT

Date: _____

* SANDWICH THEOREM :-

If $g(x) \leq f(x) \leq h(x)$

and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$

then, $\lim_{x \to a} f(x) = L$

- Questions -

Q. $\lim_{x \to 2} 5x + 2 \leq f(x) \leq x^2 + 8$

$\lim_{x \to 2} 5x + 2 \leq \lim_{x \to 2} f(x) \leq \lim_{x \to 2} x^2 + 8$

$5(2) + 2 \leq \lim_{x \to 2} f(x) \leq 2^2 + 8$

$12 \leq \lim_{x \to 2} f(x) \leq 12$

$\lim_{x \to 2} f(x) = 12$ Answer!

Date: _____

Q. $\lim_{x \to 0} -x \leq x \sin x \leq x$

$\lim_{x \to 0} -x \leq \lim_{x \to 0} x \sin x \leq \lim_{x \to 0} x$

$0 \leq \lim_{x \to 0} x \sin x \leq 0$

$\lim_{x \to 0} x \sin x = 0$ Answer!

Que. $\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right)$

$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

Multiplying by $x^2 \geq 0$

$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$

$\lim_{x \to 0} -x^2 = 0$, $\lim_{x \to 0} x^2 = 0$

Now,

$\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ Answer!

Date:

PRACTICE QUESTIONS FOR LIMITS :-

Que: Find $\lim_{x \to 3} (2x+5)$

$\lim_{x \to 3} (2x+5) = 2(3) + 5 = 6 + 5 = 11$ Ans!

Que: If $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$,

find: $\lim_{x \to a} (3f(x) - 4g(x))$

using 2nd law of limit

$3L - 4M$ Ans!

Que: Evaluate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$

$$\frac{x^2 - 4}{x - 2} = \frac{(x-2)(x+2)}{(x-2)}$$

$$= x + 2$$
$$= 2 + 2$$
$$= 4 \quad \text{Ans!}$$

MIGHTY PAPER PRODUCT

Date: _____

Que. Find left hand limit and right hand limit of the function

$$\begin{cases} x^2 & x < 1 \\ 2x + 1 & x \geq 1 \end{cases}$$

solving at $x = 1$

L.H.L = $\lim_{x \to 1^-} x^2 = 1^2 = 1$

R.H.L = $\lim_{x \to 1^+} 2(1) + 1 = 3$

Que. Determine the upper limit and lower limit of the function $f(x) = \sin \frac{1}{x}$ as $x \to 0$

$\sin \frac{1}{x}$ oscillates between -1 and 1 infinitely near 0.

- upper limit = 1
- lower limit = -1

Que. Show that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

- for $x \to 0^+$, $\frac{|x|}{x} = \frac{x}{x} = 1$
- for $x \to 0^-$, $\frac{|x|}{x} = \frac{-x}{x} = -1$

L.H.L $\neq$ R.H.L $\Rightarrow$ limit does not exist $\Leftarrow$

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Date:

Que. Evaluate $\lim_{x \to 0} \frac{\sin x}{x}$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Que. Evaluate $\lim_{x \to 2} \frac{x^2 - 3x + 2}{x - 2}$

$$\frac{(x-1)(x-2)}{(x-2)} = x - 1 = 2 - 1 = 1$$

Ans!

Que. Evaluate $\lim_{x \to 0} \frac{1}{x^2}$

for $x \to 0^+$, $\frac{1}{x^2} \longrightarrow +\infty$

for $x \to 0^-$, $\frac{1}{x^2} \longrightarrow +\infty$

Does not exist because limit tends to infinity.

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Date: _____

- By miss -

Que. $\lim_{x \to 0} \frac{\tan(x)}{x}$

$\lim_{x \to 0} \frac{\sin x / \cos x}{x}$

$\lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} = 1$

Que. $\lim_{x \to 0} \frac{1 - \cos x}{x}$

$\lim_{x \to 0} \frac{1 - \cos x}{x} = \frac{2\sin^2(x/2)}{x}$

$= \frac{2\sin^2(x/2)}{2 \cdot x/2} = \frac{\sin^2(x/2)}{x/2}$

Let $h = \frac{x}{2}$ so, $x = 2h$. as $x \to 0$, $h \to 0$

$= \frac{\sin^2 h}{2h} = \sin h \cdot \frac{\sin h}{2h}$

Now,

$\lim_{h \to 0} \sin h = 0$, $\lim_{h \to 0} \frac{\sin h}{h} = 1$

$\lim_{x \to 0} \frac{1 - \cos x}{x} = 0 \cdot \frac{1}{2} = 0$ Ans!

Date: _____

TOPIC # 3

"CONTINUETY"

A function $f(x)$ is continuous at a point $x = a$ if:

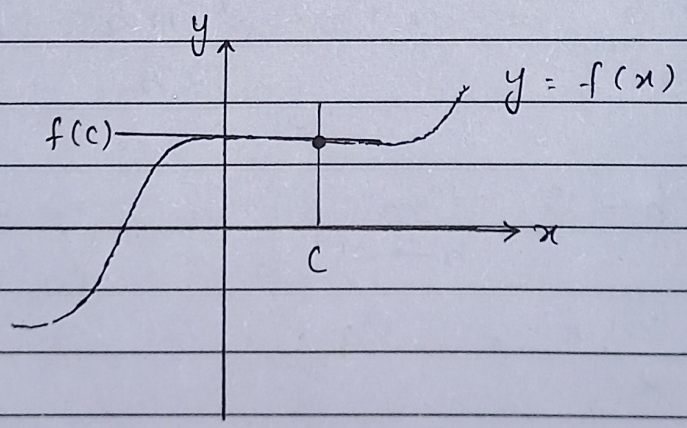
$$\lim_{x \to a} f(x) = f(a)$$

This means 3 conditions must be true

1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exists
3. $\lim_{x \to a} f(x) = f(a)$

- The function f is continuous at $x = c$

$$\lim_{x \to c} f(x) = f(c)$$



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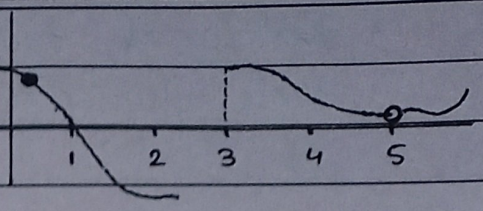
It shows there is discontinuety at $a = 1$. f is discontinuous at 1, $f(1)$ is not defined.
$f(3)$ is defined but $\lim_{x \to 3} f(x)$, does not exist.
function is discontinuous.
there $f(5)$ is defined, $\lim_{x \to 5} f(x)$ exist. f is discontinuous at 5.



$$f(x) = \begin{cases} 2x+3 & \text{for } x > 1 \\ 5 & \text{for } x = 1 \\ 6x-1 & \text{for } x < 1 \end{cases}$$

find continuely or discontinuely at $x = 1$.

- The left hand limit of $f(x)$ at $x = 1$ is

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (6x-1) = \lim_{x \to 1^-} (6x-1)$$

$$= 6(1) - 1$$

$$= 5$$

- The Right hand limit of $f(x)$ at $x = 1$ is

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} (2x+3)$$

$$= 2(1) + 3$$

$$= 5$$

- $f(1) = 5$

$L.H.L = R.H.L = f(1) = 5$

It is a continuous function

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* Rational functions are continuous in their domain.

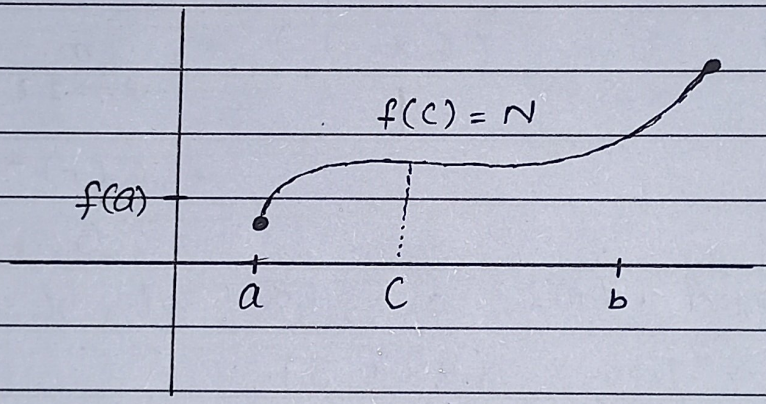
Q. $f(x) = \frac{x+1}{x-1}$

$f(1) = \frac{1+1}{1-1} = 0$ = undefined at 1

function is discontinuous!

* THE INTERMEDIATE THEOREM:-

Suppose $f(x)$ is continuous ~~ab~~ (a, b) and $f(a) < N < f(b)$, then there exist an element $(f(a,b))$ such that $f(c) = N$



Date :

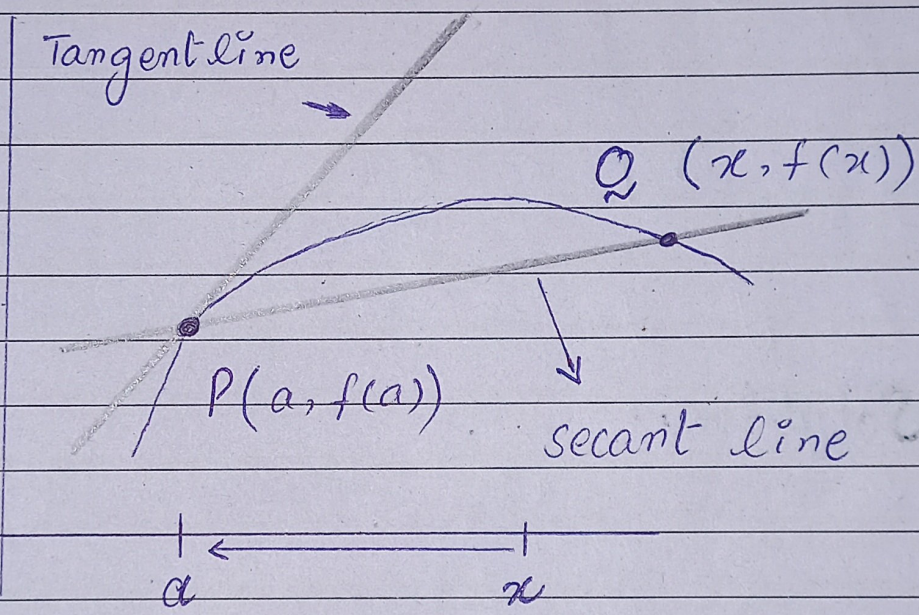
Derivative Of A Function at a Point :-

The derivative of a function ~~at a point~~ $y = f(x)$ at a point $x = a$ is defined as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

* rate of change $\frac{dy}{dx}$ = slope

motivational



Slope of secant line $\overline{PQ} = \frac{f(x) - f(a)}{x - a}$

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Note that $x \to a$ the secant line are approaching to the tangent line.

$$\frac{f(x) - f(a)}{x - a} \longrightarrow m_{PQ} = f'(a)$$

$$f'(a) = m_{PQ} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$f'(a)$ is the slope of tangent line. at $x = a$

{H. W}

- Find an equilibrium of the tangent line to the curve $y = \frac{\sqrt{x}}{1 + x^2}$

Slope : $\frac{dy}{dx}$

$$y - y_1 = m (x - x_1)$$

Solution:-

So, a horizontal tangent usually means

$$\frac{dy}{dx} = 0$$

Let

$u = \sqrt{x} = x^{1/2}$

$v = 1 + x^2$

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using quotient rule :

$$\frac{dy}{dx} = \frac{v \cdot u' - u \cdot v'}{v^2}$$

* Find u' and v'

$u = x^{1/2} \Rightarrow u' = \frac{1}{2} x^{-1/2}$

$v = 1 + x^2 \Rightarrow v' = 2x$

Applying quotient rule :

$$\frac{dy}{dx} = \frac{(1+x^2)(\frac{1}{2} x^{-1/2}) - x^{1/2}(2x)}{(1+x^2)^2}$$

simplifying numerator

$(1+x^2)(\frac{1}{2} x^{-1/2}) = \frac{1+x^2}{2\sqrt{x}}$

$2x \cdot x^{1/2} = 2x\sqrt{x}$

the whole numerator is becomes

$$\frac{1+x^2}{2\sqrt{x}} - 2x\sqrt{x}$$

Now,

$$\frac{dy}{dx} = \frac{\frac{1+x^2}{2\sqrt{x}} - 2x\sqrt{x}}{(1+x^2)^2}$$

Finding equilibrium (slop = 0)

$$\frac{1+x^2}{2\sqrt{x}} - 2x\sqrt{x} = 0$$

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multiplying both sides by $2\sqrt{x}$

"it'll remove denominator"

$$\Rightarrow (1+x^2) - 4x^2 = 0$$

$$\Rightarrow 1 + x^2 - 4x^2 = 0$$

$$\Rightarrow 1 - 3x^2 = 0$$

$$\Rightarrow x^2 = \frac{1}{3}$$

$$\Rightarrow x = \frac{1}{\sqrt{3}}$$

$\therefore$ x-coordinate where slope is 0.
i.e. tangent is horizontal.

put $x = \frac{1}{\sqrt{3}}$ in $\frac{\sqrt{x}}{1+x^2}$

$$\sqrt{x} = \sqrt{\frac{1}{\sqrt{3}}} = \frac{1}{\sqrt[4]{3}}$$

$$x^2 = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

$$y = \frac{\frac{1}{\sqrt[4]{3}}}{1+\frac{1}{3}} = \frac{1}{\sqrt[4]{3}} \cdot \frac{1}{4/3} = \frac{3}{4\sqrt[4]{3}}$$

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So,

$$\left(\frac{1}{\sqrt{3}}, \frac{3}{4\sqrt[4]{3}}\right)$$

$$y - y_1 = m(x - x_1)$$

Here

- $m = 0$ (horizontal tangent)
- $(x_1, y_1) = \left(\frac{1}{\sqrt{3}}, \frac{3}{4\sqrt[4]{3}}\right)$

So,

$$y - \frac{3}{4\sqrt[4]{3}} = 0\left(x - \frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow y = \frac{3}{4\sqrt[4]{3}}$$

point where tangent is horizontal

$$\left(\frac{1}{\sqrt{3}}, \frac{3}{4\sqrt[4]{3}}\right)$$

Tangent equation;

$$y = \frac{3}{4\sqrt[4]{3}}$$

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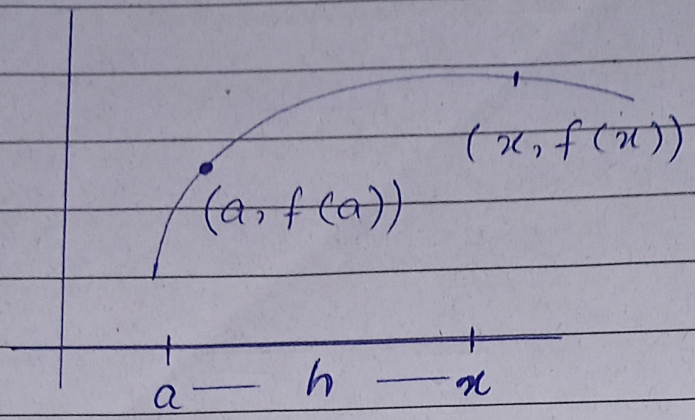
Alternative Definition :-



let $h = x - a$ then $x = a + h$
also notes $x \to a$ if $h \to 0$

therefore $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} =$

$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

If $\lim_{h \to 0} f(a+h) - f(a)$ does not exist

then we say that $f'(a)$ does not exist.
f is not differentiable at $x = a$

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1) $f(x) = x^2$ $\qquad (x+h)^2$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2hx + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{h(2x + h)}{h}$$

$$= 2x + 0$$

$$= 2x$$

Que: $f(x) = x^2 + 8x + 9$

$$\lim_{h \to 0} \frac{(x+h)^2 + 8(x+h) + 9 - (x^2 + 8x + 9)}{h}$$

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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LAWS:

(1) $[f(x) + g(x)]' = f'(x) + g'(x)$

(2) $[f(x) - g(x)]' = f'(x) - g'(x)$

PRODUCT RULE:-

* $\frac{dy}{dx}[f(x) \cdot g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x)$

QUOTIENT RULE:-

* $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\, d/dx f(x) - f(x)\, d/dx\, g(x)}{[g(x)]^2}$

* $\frac{d}{dx}(c) = 0$
* $\frac{d}{dx}(x^n) = nx^{n-1}$

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Homework Questions :-

1) $f(x) = (x^3+1)(x^2)$

using Product rule

$\because \frac{d}{dx}(u \cdot v) = u'v + uv'$

$u = (x^3+1)$

$u' = 3x^2$

$v = x^2$

$v' = 2x$

apply u' and v' on product rule.

$f'(x) = (3x^2)(x^2) + (x^3+1)(2x)$

$f'(x) = 3x^4 + 2x(x^3+1)$

$f'(x) = 3x^4 + 2x^4 + 2x$

$f'(x) = 5x^4 + 2x$ Answer!

2) $f(x) = \frac{x^3+1}{x^4-6}$

using quotient rule.

$\because \frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$

$u = x^3 + 1$

$u' = 3x^2$

$v = x^4 - 6$

$v' = 4x^3$

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applying in the formula.

$$f'(x) = \frac{(3x^2)(x^4-6) - (x^3+1)(4x^3)}{(x^4-6)^2}$$

$$f'(x) = \frac{3x^2(x^4-6) - 4x^3(x^3+1)}{(x^4-6)^2}$$ Answer!

3) $f(x) = x^{2/3}$

using power rule :

$\because \frac{d}{dx}[x^n] = nx^{n-1}$

so,

$$f'(x) = \frac{2}{3} x^{-1/3} \text{ or } \frac{2}{3\sqrt[3]{x}}$$

Answer!

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Question: $f(x) = \sin(x^2)$

$$f(x) = \sin(x^2)$$

$$\Rightarrow \frac{d}{dx} \sin(x^2)$$

$$\Rightarrow \frac{d}{dx} \sin x^2 \cdot \frac{d}{dx} x^2$$

$$\Rightarrow \cos x^2 \cdot 2x$$

$$\Rightarrow 2x \cos x^2$$ Answer!

Question: $y = [\sin(x)]^2$

$$\frac{dy}{dx} = \frac{d}{dx} [\sin(x)]^2$$

$$= 2 \sin x \frac{d}{dx} \sin x \cdot \frac{d}{dx} x$$

$$= 2 \sin x \cos x$$ Answer!

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Date :

Power Rule With Chain Rule:

$$\frac{d}{dx}(x^n) = n x^{n-1} \quad \text{(Power Rule)}$$

$$\frac{d}{dx}[f(x)]^n = n(f(x))^{n-1} \cdot f'(x)$$

Question: $y = \sqrt{\sec(x^3)}$

$$y = \sqrt{\sec(x^3)}$$

$$\frac{dy}{dx} = [\sec(x^3)]^{1/2}$$

$$\frac{d}{dx}[\sec(x^3)] = \sec(x^3)\tan(x^3) \cdot 3x^2$$

Applying chain rule again

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\sec(x^3)}} \cdot \sec(x^3)\tan(x^3) \cdot 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2 \sec(x^3)\tan x^3}{2\sqrt{\sec(x^3)}}$$

$$f'(x) = \frac{3x^2 \sec(x^3)\tan(x^3)}{2\sqrt{\sec(x^3)}}$$

Answer!

M.M

Date :

Question : $f(x) = \sin(\cos(\tan(x)))$

Let,

$u = \tan(x)$

$v = \cos(u)$

$f = \sin(v)$

Now,

$$f'(x) = \cos(\cos(\tan(x))) \cdot (-\sin(\tan(x))) \cdot \sec^2 x$$

$$f'(x) = -\cos(\cos(\tan x)) \cdot \sin(\tan x) \cdot \sec^2 x$$

Answer!

Question : $f(x) = \sin x^2$

$$\frac{d}{dx} \sin x^2 = \cos x^2 \frac{d}{dx} x^2$$

$$f'(x) = 2x \cos x^2$$ Answer!

Question : $f(x) = (\sin x)^2$

$$\frac{d}{dx} (\sin x)^2 = 2 \sin x \cdot \frac{d}{dx} (\sin x)$$

$$f'(x) = 2 \sin x \cdot \cos x$$ Answer!

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Date :

Question : $f(x) = \dfrac{1}{3\sqrt{x^2}}$

$$f'(x) = \frac{1}{3} \cdot (-2x^{-3})$$

$$f'(x) = \frac{-2}{3x^3}$$

Answer !

Question : $y = \sqrt{x(1+x^2)}$

chain Rule + product rule

$$y = [x(1+x^2)]^{1/2}$$

let $u = x(1+x^2)$

$$u' = (1)(1+x^2) + x(2x)$$

$$u' = 1 + x^2 + 2x^2$$

$$u' = 1 + 3x^2$$

Now,

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x(1+x^2)}} \cdot (1 + 3x^2)$$

$$f'(x) = \frac{1 + 3x^2}{2\sqrt{x(1+x^2)}}$$

M.M

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Date :

Question : $f(x) = \dfrac{\sec x}{1+\tan x}$

using quotient rule

let,

$u = \sec x$

$u' = \sec x \tan x$

$v = 1 + \tan x$

$v' = \sec^2 x$

applying quotient rule.

$$f'(x) = \frac{\sec x \tan x (1+\tan x) - \sec x (\sec^2 x)}{(1+\tan x)^2}$$

Answer!

M.M

Date: _____

* Implicit Function :-

$$\frac{d}{dx}(y)^n = n y^{n-1} \frac{dy}{dx}$$

$$\therefore \frac{d}{dx}[g(y)] = g'(y) \cdot \frac{dy}{dx}$$

Q. $x^2 + y^2 = 1$ find $\frac{dy}{dx}$

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y} \quad \text{Ans!}$$

Q. $x^2 y^3 = x$

$$y^3 (x^2)' + x^2 (y^3)' = 1$$

$$2x y^3 + 3x^2 y^2 \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1 - 2x y^3}{3x^2 y^2}$$

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Date:

Q. $x^3 + y^3 = 6xy$

$$3x^2 + 3y^2 \frac{dy}{dx} = 6[x'y + y'x]$$

$$3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$$

$$3y^2 \frac{dy}{dx} - 6x \frac{dy}{dx} = 6y - 3x^2$$

$$\frac{dy}{dx}[3y^2 - 6x] = 6y - 3x^2$$

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x}$$

$$\frac{dy}{dx} = \frac{3(2y - x^2)}{3(y^2 - 2x)}$$

$$\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$$ Answer!

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Q. $\sin(y^2) = x^2$

$$\cos y^2 \cdot 2y \frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{2x}{\cos y^2 \cdot 2y}$$

$$\frac{dy}{dx} = \frac{x}{y \cdot \cos y^2}$$ Ans!

* Derivative Of Inverse Function Using Implicit Function :-

(Implicit differentiation)

Que. $y = \sin^{-1}(x)$

$$x = \sin y$$

$$\frac{d}{dx} x = \frac{d}{dx} \sin y$$

$$1 = \cos y \left(\frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

Date: _____

$\because \sin^2 y + \cos^2 y = 1$

$x^2 + \cos^2 y = 1$

$\because \cos(y) = \sqrt{1-x^2}$

$$\frac{dy}{dx} = \frac{1}{1-x^2}$$ Answer!

Q. $y = \tan^{-1} x$

$\tan y = x$

$\sec^2 y \cdot \frac{dy}{dx} = 1$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$\because \tan y = x$

$\sec^2 y = 1 + x^2$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$ Ans!

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Que. $y = \ln(x)$

$$e^y = x$$

$$\frac{d}{dx} e^y = \frac{d}{dx} x$$

$$e^y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{e^y}$$

$\because e^y = x$

$$\frac{dy}{dx} = \frac{1}{x}$$

Answer!

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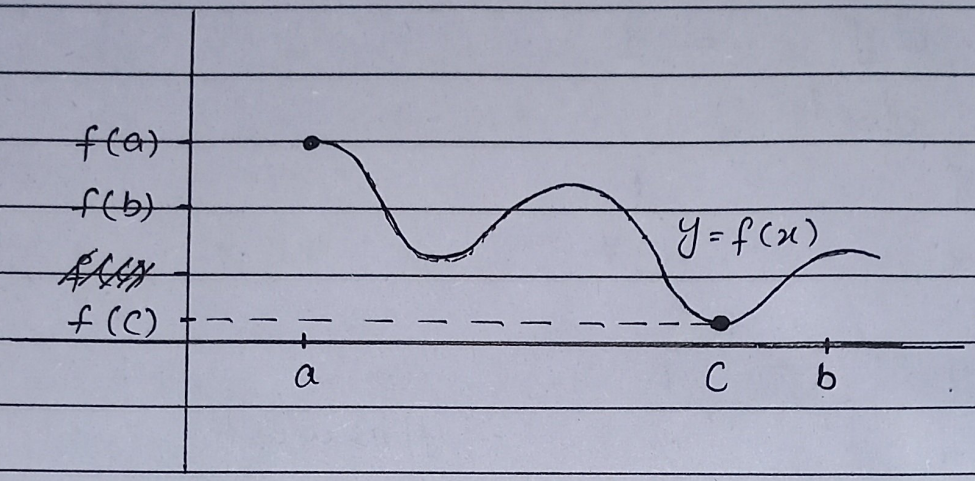
* Absolute Maximum & Minimum Value :-



- **Absolute Maximum :**
  The highest value of a function on a given interval.

- **Absolute Minimum :**
  The lowest value of a function on a given interval.

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* CRITICAL POINT:-

A point "x = c" is called a critical point of $y = f(x)$ if $f'(c) = 0$, or $f''(c)$ is not differentiable at $x = c$.

Examples:-

Q. find the critical point of $f(x) = x^3 - 3x^2 + 1$

* A point c is critical if $f'(c) = 0$

$$f'(x) = 3x^2 - 6x$$
$$f'(c) = 3c^2 - 6c$$
$$3c^2 - 6c = 0$$
$$3c(c-2) = 0$$
$$c - 2 = 0$$

critical points = $c = 0$, $c = 2$

Q. $f(x) = x^{3/5}$

$$f'(x) = \frac{3}{5} x^{3/5 - 1}$$
$$= \frac{3}{5} x^{-2/5}$$
$$f'(x) = \frac{3}{5x^{2/5}}$$

∵ $f'(0)$ is not defined & it is not differentiable at "0" so "0" is a critical point.

Date: _____

Que. Find the absolute maximum and minimum of $f(x) = x^2 - 3x^2 + 1$ on $[\underset{a}{-1/2}, \underset{b}{4}]$

Step : 1

$f(a) = f(-1/2) = 1/8$

$f(b) = f(4) = 17$

Step : 2

finding critical point

$f'(x) = 3x^2 - 6x$

$f'(c) = 3c^2 - 6c$

$3c^2 - 6c = 0$

$c = 0 \quad , \quad c = 2$

Step : 3

$f(0) = 1 \quad (x^3 - 3x^2 + 1)$

$f(2) = -3 \quad (x^3 - 3x^2 + 1)$

Step : 4

Compare $f(a)$, $f(b)$, $f(c)$

$f(-1/2) = 1/8$

$f(4) = 17 \rightarrow$ Absolute maximum

$f(2) = -3 \rightarrow$ Absolute minimum

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Date: _____

Que. $f(x) = x^{3/5}$ on $[-1, 1]$. Find absolute maximum and minimum.

Step: 1

$f(a) = x^{3/5} = (-1)^{3/5} = -1$

$f(b) = x^{3/5} = (1)^{3/5} = 1$

Step: 2 critical point

$$f'(x) = \frac{3}{5x^{2/5}}$$

$$f'(c) = \frac{3}{5c^{2/5}}$$

$c = 0$

Step: 3

$f(0) = 0^{3/5} = 0$

Step: 4

Compare $f(a)$, $f(b)$, $f(c)$

$f(-1) = -1 \rightarrow$ min.

$f(1) = 1 \rightarrow$ max.

$f(0) = 0$

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Date: _____

Que. $f(x) = \sin(x)$ on $[0, 2\pi]$

Step : 1

$f(a) = f(0) = \sin(0) = 0$

$f(b) = f(2\pi) = \sin(2\pi) = 0$

Step : 2

$f'(x) = \cos x$

$f'(c) = \cos(c)$

$\cos(c) = 0$

$c = \cos^{-1} 0$

$c = \frac{\pi}{2}, \quad \frac{3\pi}{2}$

Step : 3

$f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$

$f\left(\frac{3\pi}{2}\right) = \sin \frac{3\pi}{2} = -1$

Step : 4 comparing

$f(0) = 0$

$f(2\pi) = 0$

$f(\pi/2) = 1 \rightarrow$ max.

$f(3\pi/2) = -1 \rightarrow$ min.

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